AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous by default. Supervisable by design.

ROBUST SUPERVISABLE BROAD SEARCH

ROADMAP / TALK ORDER

The run, in five moves

A single route from the problem to the proof, then the product experience.

01

PROBLEM

Why current agents fail

Set the task and its failure modes.

02

SYSTEM

A system for sustained research

Keep evidence while resetting the scientist.

03

EVALUATION

From claim to evidence

Measure what survives the run.

04

INSIGHTS

What the run taught us

Depth, breadth, recovery, and review.

01 / PROBLEM · THE RESEARCH TASK

Automating ML experimentation

Given a dataset and a scoring metric, an agent must build a model and improve it through repeated experiments.

01	02	03	04	05
Read the problem	**Inspect data**	**Engineer features**	**Train + tune**	**Evaluate**
What should the model predict?	What patterns does the data contain?	Turn raw data into useful model inputs	Fit a model and adjust its settings	Check performance on validation data

REFLECT + REVISE

The agent uses the results to choose what to change, revises the code, and runs the next experiment.

THE AGENT'S JOB

Write the code and run the full loop autonomously.
This includes building the pipeline and deciding which experiments to try next.

THE CHALLENGE GOAL

Reproduce, then improve on the organizer's baseline.
Develop with training data and public validation. The organizer scores the final submission on a hidden test set.

01 / PROBLEM · FAILURE MODES

Why current agents fail

01 WORKFLOW

Observe → Code → **Error** → **Recover**

Coding Harness

Runtime Tools

Acquire missing tools at runtime.

- ✓ Resume after unexpected errors.
- ✓ Extend with Skills & MCP.

02 SINGLE AGENT

last frame — next move

history-shaped

single path

Fresh reasoning. Audited evidence.

- ✓ Reopen search.
- ✓ Check the science.

03 TREE SEARCH

READ ONCE → BRANCH EDITS

Persist the read. Vary the hypothesis.

Reuse context across experiments.

- ✓ Raise task granularity.
- ✓ Parallelize across branches.

- Adaptive tools
- Fresh reasoning
- Audited evidence
- Reusable context

Build the system →

02 / SYSTEM · DESIGN

A system for sustained research

Environment

- Task contract
- Curated data + evaluator
- Starter baseline
- Budget + permissions

Research World

A · VERIFIED / EVIDENCE

- Verified knowledge
- Evidence ledger

B · CURATED / REVISABLE

- Research brief
- Working priors

C · IMPLEMENTATION

- Implementation State

What persists across trajectories

CONTEXT RESET

Fresh Scientist

Scientist

owns scientific judgment

- reads the research world
- forms hypotheses
- runs experiments
- interprets evidence
- writes a report

inherits experience, not momentum

META Review

META

owns what survives

- audits claim ↔ evidence validity
- scopes conclusions
- maintains shared research memory
- crystallizes retained implementation into State

Selective persistence / trajectory handoff

Coding-agent harness + runtime

- Workspace isolation
- Runner + evaluator
- Logging
- Artifact capture
- State operations

BACK TO HAND ×

We are not tokenmaxxing

PROJECT-LEVEL RESOURCE ACCOUNTING

~$10

TOTAL PROJECT COST

Subscription-equivalent usage from first line of code to final result.

coding · debugging · agent loops · all 11 experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire autonomous research loop.

HARDWARE

1 × MacBook M2

All development, feature engineering, and validation ran on one consumer laptop.

COMPUTE

0 GPU-hours

100% CPU NumPy training and inference; zero cloud GPU clusters required.

> Performance came from the autonomous research loop, not an expensive compute budget.

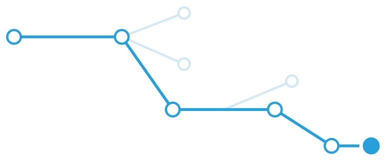
04 / INSIGHTS · MODEL DIVERSITY

Model diversity improves the result.

Cycling GPT and Gemini brings different search styles into one research world.

Deepen an idea

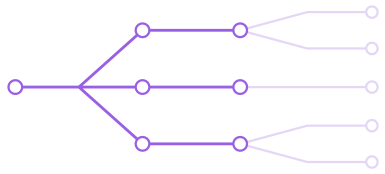
GPT 5.6 Luna



Refine the current approach.
Follow promising details further.

Open new directions

Gemini 3.7 Flash



Try alternative approaches.
Challenge the current direction.

Model diversity improved the retained result.

GPT → Gemini → GPT → Gemini

Shared research memory · Gemini META

Public-validation Primary

Gemini-only	GPT + Gemini
0.6052	**0.6059363**

04 / INSIGHTS · PARALLEL RESEARCH

Explore in parallel. Build on the findings.

One branch continues. Another contributes a useful perspective.

A ≈0.6048
Redundant findings

B ≈0.6048
Chosen for further work

C ≈0.6039
Different modeling ideas

Reviewer

Compare results and research quality.

Fresh Scientist

Develop branch B, informed by C's findings.

0.6055536

Public-validation Primary

A lower-scoring branch can still contribute an idea.

The new model used pairwise ranking, higher rank and rank-normalized ensembling.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS

Give every agent a branch.

main

clean trunk

- agent A / branch
- agent B / branch
- agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP

Leave the next move in GitHub.

01 Issue / prompt — inject the task →

02 GitHub Action — run the agent →

03 Commit — review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

PRACTICAL WORKFLOW Recommended engineering pattern for unattended overnight exploration.

05 / USER EXPERIENCE

Autonomous by default.
Supervisable by design.

Research is an open-world task.
We need an open-world coding agent.

01 OPEN ACTION SPACE

- Shell
- Files
- Browser
- Git

Use what exists

02 SELF-RECOVERY

- Read errors
- Inspect docs
- Inspect source
- Install packages

Fix what breaks

03 RUNTIME EXTENSION

- MCP
- Skills
- CLI
- Packages

Add what's missing

APPENDIX / THE TEAM

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD

CZ

Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture · integration

ZZ

Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles · evidence review

SX

Shilin Xu

DATA / METRICS

Data workflow · contracts · alignment

GG

GE GAO

RUNTIME / RELIABILITY

Execution · testing · integration support

JL

Jiran Li

EVIDENCE / REPRODUCIBILITY

Telemetry · documentation · packaging

Every handoff kept the research world moving.